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WeMM-Embedding-9B

Hugging Face | Technical Report

GitHub WeMM-Embedding

WeMM-Embedding-9B is a universal multimodal embedding model built on Qwen3.5. It accepts text, images, videos, visual documents, and interleaved multimodal inputs, and returns a 4,096-dimensional L2-normalized embedding. Audio input is not supported.

Installation

```
pip install torch transformers==5.2.0 "qwen-
  "sentence-transformers>=5.7.0" "accelerate
```

Transformers

```
import torch
from qwen_vl_utils import process_vision_inf
from transformers import AutoModel, AutoProc
```

Downloads last month
385

Safetensors

Model size | 9B params

Tensor type | BF16

Chat template

Files info

Inference Providers NEW

Feature Extraction

This model isn't deployed by any Inference Provider.

Ask for provider support

Model tree for tencent/WeMM-Embe...

Base model Qwen/Qwen3.5-9B-Base

Finetuned Qwen/Qwen3.5-9B

Finetuned (670) .... this model

Space using tencent/WeMM-Embe... 1

Mike0021/WeMM-Embedding-9B

```python
model_id = "tencent/WeMM-Embedding-9B"
processor = AutoProcessor.from_pretrained(mo
model = AutoModel.from_pretrained(
    model_id, trust_remote_code=True, dtype=
).cuda().eval()

messages = [{"role": "user", "content": [
    {"type": "image", "image": "/path/to/ima
    {"type": "video", "video": "/path/to/vid
    {"type": "text", "text": "This can be an
]}]
text = processor.apply_chat_template(
    messages, tokenize=False, add_generation
)
images, videos, video_kwargs = process_visio
    messages,
    image_patch_size=16,
    return_video_kwargs=True,
    return_video_metadata=True,
)
if videos is not None:
    videos, video_metadata = zip(*videos)
    videos, video_metadata = list(videos), l
else:
    video_metadata = None
inputs = processor(
    text=text,
    images=images,
    videos=videos,
    video_metadata=video_metadata,
    return_tensors="pt",
    **video_kwargs,
).to("cuda")

with torch.inference_mode():
    embedding = model.embedding(**inputs)
```

Use any subset of the content items to encode text, image, or video independently.

Collection including tencent/WeMM...

WeMM-Embedding Collection
4 items • Updated 3 days ago • 21

Paper for tencent/WeMM-Embeddin...

WeMM-Embedding: WeChat Multi-...
Paper • 2608.24053 • Publi... • 63

Sentence Transformers

```
from sentence_transformers import SentenceTra

model_id = "tencent/WeMM-Embedding-9B"
model = SentenceTransformer(model_id, trust_

queries = [
    "Which Llama 4 model variants are availal
    "How is mapo tofu prepared?",
]
documents = [
    "Mapo tofu is a Sichuan dish of soft tofu
    {
        "image": "https://huggingface.co/data
        "text": "Represent this image.",
    },
    {
        "video": "https://huggingface.co/data
        "text": "Represent this video.",
    },
]

query_embeddings = model.encode_query(queries
document_embeddings = model.encode_document((
print(query_embeddings.shape, document_embedd
# (2, 4096) (3, 4096)

similarities = model.similarity(query_embeddi
print(similarities)
# tensor([[0.2153, 0.5843, 0.1221],
#         [0.7665, 0.2604, 0.5366]])
```

Each input is a string, a URL or path, a `PIL.Image`, or a dict combining `image`, `video`, and `text` keys. Put `image` or `video` before `text` so the prompt matches the ordering used above. Chat messages such as `{"role": "user", "content": [{"type": "image", "image": ...}, {"type": "text", "text": ...}]}`

are also accepted, which is the way to interleave several images or videos in one input.

Matryoshka Embeddings

```
d = 256
embedding_d = torch.nn.functional.normalize(
```

With Sentence Transformers, pass `truncate_dim` and let it renormalize:

```
embeddings_d = model.encode_document(documen
```

Use a dimension listed in `model.config.matryoshka_dimensions`.

Serving

vLLM `0.27.0`:

```
MODEL_PATH=/path/to/WeMM-Embedding-9B
vllm serve "$MODEL_PATH" \
  --runner pooling \
  --chat-template "$MODEL_PATH/embedding_cha
```

SGLang `0.5.9`:

```
MODEL_PATH=/path/to/WeMM-Embedding-9B
python patch_sglang_video.py
python -m sglang.launch_server \
  --model-path "$MODEL_PATH" \
  --is-embedding \
  --enable-precise-embedding-interpolation
```

Evaluation

MMEB-v2

Results on 78 datasets from Table 1 of the technical report. Image and video tasks use Hit@1, while visual-document tasks use NDCG@5. Higher is better.

Model	Size	AVG	Image	Video	VisDoc
VLM2Vec	2B	47.8	59.7	29.0	44.0
GME	2B	55.4	51.9	33.9	76.8
VLM2Vec-V2	2B	59.3	64.9	34.9	69.2
Qwen3-VL-Embedding	2B	73.2	75.0	61.9	79.2
DME-Small†	2B	74.8	75.9	65.6	79.9
WeMM-Embedding	**2B**	**77.9**	**79.6**	**70.8**	**80.7**
WeMM-Embedding	**4B**	**79.2**	**80.8**	**72.1**	**82.0**
VLM2Vec	8B	53.2	65.5	34.0	49.1
GME	8B	59.2	56.0	38.6	79.3
Qwen3-VL-Embedding	8B	77.8	80.1	67.1	82.4
DME-Medium†	9B	78.4	79.8	70.8	82.0
WeMM-Embedding	**9B**	**80.6**	**81.9**	**74.3**	**83.3**

† Closed-source leaderboard submission without publicly released model weights or a public inference endpoint.

MMEB-v3

Results on all 190 tasks from Table 2 of the technical report. V3-All includes the 78 MMEB-v2 tasks, 53 text tasks, 47 agent tasks, 11 audio tasks, and MCMR. Unsupported tasks are assigned a score of zero.

Model	Size	V3-All	Text	Agent	MCMR	Aud
VLM2Vec-V2	2B	38.3	24.5	28.7	4.1	0.0
Omni-Embed-Nemotron	3B	43.5	39.2	36.5	26.1	36.5
E5-Omni	3B	44.6	26.7	36.9	31.9	30.8
Qwen3-VL-Embedding	2B	50.9	39.2	39.3	42.0	0.0
WeMM-Embedding	**2B**	**56.0**	**45.3**	**45.1**	**42.5**	**0.0**
WeMM-Embedding	**4B**	**58.2**	**47.9**	**49.0**	**41.9**	**0.0**
WAVE	7B	26.3	13.7	11.3	8.9	31.8
VLM2Vec	8B	32.9	22.2	19.7	0.9	0.0
LCO-Embedding-Omni	7B	40.6	32.4	27.8	20.0	43.2
GME	8B	43.6	37.1	35.6	27.3	0.0
E5-Omni	7B	47.1	26.9	36.7	41.1	43.0
Tianmu-Emb-Uni	8B	53.3	43.6	39.4	38.8	38.9

Model	Size	V3-All	Text	Agent	MCMR	Aud
Qwen3-VL-Embedding	8B	53.5	42.5	38.4	38.0	0.0
WeMM-Embedding	**9B**	**59.5**	**48.8**	**51.0**	**49.3**	**0.0**

Text results use NDCG@5; agent, MCMR, and audio results use Hit@1.

Citation

If you find this repository useful, please consider giving a star ⭐ and citation

```
@article{wemm-embedding,
      title={WeMM-Embedding: WeChat Multi-Mo
      author={Junjie Zhou and Ke Mei and Lei
      year={2026},
      eprint={2608.24053},
      archivePrefix={arXiv},
      primaryClass={cs.CV},
      url={https://arxiv.org/abs/2608.24053}
}
```

License

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